DSA 8420 Spring 2025
Homework 6

Due February 25, 2025

1. Consider the following linear program.

$$
\begin{aligned}
\max \ \ z = 2x_1 - 4x_2 + 7x_3 & \\
s.t. \qquad x_1 - \ \ x_2 + \ \ x_3 &= 2 \\
x_1 + \ \ x_2 + 2x_3 &\leq 10 \\
x_1 + \ \ x_2 - \ \ x_3 &\geq 6 \\
x_1,\, x_2,\, x_3 &\geq 0.
\end{aligned}
$$

   (a) (4 points) Transform the problem into standard form.

   (b) (4 points) Construct the Phase I problem of the standard-form LP.

   (c) (20 points) Solve the problem in (a) by the two-phase method.

2. Consider the following linear program.

$$
\begin{aligned}
\max \ \ z = 2x_1 + \ \ x_2 & \\
s.t. \qquad x_1 - \ \ x_2 &\geq 8 \\
2x_1 + 3x_2 &\leq 24 \\
2x_1 + \ \ x_2 &\leq 12 \\
x_1,\, x_2 &\geq 0
\end{aligned}
$$

   (a) (4 points) Transform the problem into standard form.

   (b) (4 points) Construct the Phase I problem of the standard-form LP.

   (c) (14 points) Use the two-phase method to show that the problem is infeasible.